

GATE CSE 2026

Predicted Paper

GATE CS 2026 - Predicted Paper 3: Advanced and Emerging

Focus: Machine Learning, Computer Vision, NLP, Information Security, Ethics

Total Questions: 65 | Total Marks: 100 | Duration: 180 min

Generated by PadhaiShuru | padhaishuru.com

General Aptitude

Q1. [1 mark] Synonym of: RESILIENT

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q2. [1 mark] Sum of first 20 natural numbers?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q3. [2 marks] Despite setbacks, she remained _____.

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q4. [1 mark] Probability of at least one head in 2 coin tosses?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q5. [2 marks] Speed = 60 km/h. Distance in 2.5 hours?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q6. [1 mark] OPPOSITE of: AMELIORATE

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q7. [1 mark] If $x+y=10$, $x-y=4$, then $x = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q8. [2 marks] Grammatically correct sentence:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q9. [2 marks] Pipes A(12h), B(18h) fill tank together. Time?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q10. [1 mark] Find the odd pair: A) Cat:Kitten B) Dog:Puppy C) Cow:Cub D) Horse:Foal

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Engineering Mathematics

Q11. [1 mark] Rank of identity matrix I₃?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Identity matrix always has full rank = n.

Q12. [1 mark] Binomial: n=5, p=0.5. P(X=3) = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $C(5,3) \cdot (0.5)^3 \cdot (0.5)^2 = 10/32 = 5/16$.

Q13. [1 mark] $\lim_{x \rightarrow 0} \sin(x)/x = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Standard limit: $\lim_{x \rightarrow 0} \sin(x)/x = 1$.

Q14. [2 marks] Group axioms: closure, associativity, identity, inverse. (Z,+) is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: (Z,+) is an Abelian group: all four axioms + commutativity hold.

Q15. [2 marks] LU decomposition of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. L₁₁ = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: In LU decomposition without pivoting, $L_{11} = A_{11} = 2$.

Technical (CSE)

Q16. [1 mark] Decision tree split criterion in ID3?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: ID3 uses Information Gain (entropy reduction). CART uses Gini index.

Q17. [1 mark] Linear regression minimizes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: OLS minimizes sum of squared errors = MSE.

Q18. [2 marks] Backpropagation uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Backprop applies chain rule to compute gradients, then gradient descent to update weights.

Q19. [1 mark] K-means is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: K-means is unsupervised learning (no labels).

Q20. [2 marks] PCA finds:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: PCA projects data onto eigenvectors of covariance matrix - directions of maximum variance.

Q21. [2 marks] Naive Bayes assumes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Naive = conditional independence of features given class label.

Q22. [1 mark] High variance indicates:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: High variance = model fits training too well -> overfitting. High bias = underfitting.

Q23. [2 marks] Kernel trick in SVM:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Kernel implicitly maps data to higher-dimensional space without explicit transformation.

Q24. [2 marks] Random Forest reduces:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Bagging reduces variance by averaging multiple models. Bias remains similar.

Q25. [2 marks] Gradient descent with large learning rate:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Too large learning rate causes overshooting -> divergence.

Q26. [1 mark] RGB to Grayscale formula?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Standard luminance formula weighted by human eye sensitivity.

Q27. [1 mark] Sobel operator detects:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Sobel computes gradient magnitude -> edge detection.

Q28. [2 marks] Convolution layer purpose:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Conv layers extract hierarchical features: edges -> textures -> patterns.

Q29. [1 mark] Image histogram shows:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Histogram = distribution of pixel intensity values.

Q30. [2 marks] YOLO architecture is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: YOLO = single-stage detector. Faster R-CNN is two-stage.

Q31. [1 mark] Max pooling purpose:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Pooling downsamples feature maps -> reduces spatial dimensions.

Q32. [2 marks] Semantic vs Instance segmentation:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Semantic: pixel-wise class. Instance: separate objects of same class (different colors).

Q33. [2 marks] Vision Transformer (ViT) uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: ViT splits image into patches, applies transformer self-attention.

Q34. [1 mark] Tokenization is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Tokenization splits raw text into words/subwords/tokens.

Q35. [1 mark] Word2Vec uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Word2Vec uses shallow neural networks: CBOW or Skip-gram.

Q36. [2 marks] Transformer self-attention computes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Attention: $Q \cdot K^T / \sqrt{d}$. V. Each token learns what to attend to.

Q37. [1 mark] POS tagging assigns:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: POS = Noun/Verb/Adjective/etc. per word.

Q38. [2 marks] NER identifies:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: NER extracts named entities: persons, organizations, locations, dates.

Q39. [2 marks] BERT is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BERT = Bidirectional Encoder Representations from Transformers. Masked LM pretraining.

Q40. [1 mark] Bag of Words represents text as:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BoW = vector of word frequencies, ignoring order.

Q41. [2 marks] In attention, Query comes from:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Q represents the current token asking what to attend to. K,V can be from encoder or decoder.

Q42. [1 mark] Symmetric key uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Symmetric = same key for encrypt/decrypt (AES, DES).

Q43. [2 marks] RSA key generation uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: RSA: choose p, q (primes) $\rightarrow n=pq \rightarrow \phi(n)=(p-1)(q-1) \rightarrow$ choose e, d .

Q44. [1 mark] Digital signature provides:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Signatures verify sender (auth) and detect tampering (integrity). Not confidentiality.

Q45. [2 marks] HTTPS uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: HTTPS = HTTP over TLS/SSL for encrypted communication.

Q46. [1 mark] Man-in-the-middle attack:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: MITM attacker intercepts and may alter communication between parties.

Q47. [2 marks] Firewall operates at:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Firewalls can operate at network (packet filter), transport (stateful), or application (proxy) layer.

Q48. [2 marks] SHA-256 produces:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SHA-256 produces 256-bit hash. SHA-1=160, SHA-512=512, MD5=128.

Q49. [1 mark] Software patent protects:

- @. [object Object]
- A. [object Object]

- B. [object Object]
- C. [object Object]

Explanation: Patents protect novel, non-obvious ideas/algorithms. Copyright protects expression.

Q50. [2 marks] Right to be forgotten relates to:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Right to be forgotten = data privacy - individuals can request data removal.

Q51. [1 mark] Algorithmic bias means:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Bias = systematic prejudice in outcomes due to biased training data or design.

Q52. [1 mark] ACM Code addresses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: ACM Code of Ethics covers professional conduct, responsibility, privacy, etc.

Q53. [2 marks] Internet censorship debate centers on:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Core tension: freedom of expression vs preventing harm (hate speech, illegal content).

Q54. [2 marks] Trolley problem in autonomous vehicles relates to:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Trolley problem = ethical dilemma about harm minimization in unavoidable crash scenarios.

Q55. [1 mark] Most permissive open source license:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: MIT/BSD are most permissive (public domain-like). GPL is copyleft.

Q56. [1 mark] Full binary tree with n internal nodes has:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Full binary tree: n internal $\rightarrow$ n+1 leaves. Total nodes = 2n+1.

Q57. [2 marks] Inorder traversal of BST gives:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Inorder of BST = elements in sorted (ascending) order.

Q58. [2 marks] Load factor $\alpha = n/m$. $n=50$, $m=25$. $\alpha = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $\alpha = n/m = 50/25 = 2$.

Q59. [1 mark] Circular linked list last node points to:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Circular LL: last.next = head. No NULL termination.

Q60. [1 mark] Stack insertion/removal:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Stack = LIFO. Both push and pop at top (one end).

Q61. [2 marks] Binary tree height h has max nodes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Perfect binary tree at height h (levels) has $2^h - 1$ nodes.

Q62. [2 marks] Min-heap property:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Min-heap: parent value $\leq$ child values.

Q63. [1 mark] Queue follows:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Queue = FIFO (First In First Out).

Q64. [2 marks] Topological sort requires:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Topological sort only defined for DAG (Directed Acyclic Graph).

Q65. [2 marks] 0/1 Knapsack is solved by:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 0/1 Knapsack is classic DP. Greedy works for fractional but not 0/1.

Answer Key

General Aptitude

Q1: B
Q2: C
Q3: A
Q4: C
Q5: B
Q6: B
Q7: C
Q8: C
Q9: A
Q10: C

Engineering Mathematics

Q11: C
Q12: B
Q13: B
Q14: C
Q15: A

Technical (CSE)

Q16: B
Q17: B
Q18: B
Q19: B
Q20: B
Q21: B
Q22: B
Q23: B
Q24: B
Q25: B
Q26: B
Q27: B
Q28: B
Q29: B
Q30: B
Q31: B
Q32: B
Q33: B
Q34: B
Q35: B
Q36: B
Q37: B
Q38: B
Q39: B
Q40: B
Q41: B
Q42: B
Q43: A
Q44: B
Q45: B
Q46: B
Q47: B
Q48: B
Q49: B
Q50: B
Q51: B
Q52: B
Q53: B
Q54: B
Q55: B
Q56: B
Q57: B
Q58: B
Q59: B
Q60: C
Q61: C
Q62: B

Q63: B
Q64: B
Q65: B